



FRACTAL ANTENNA DESIGN FOR WIRELESS POWER TRANSFER APPLICATION



A PROJECT REPORT

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ABSTRACT

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Wireless Power Transfer (WPT) systems require compact, efficient, and high-performance antennas to transmit energy effectively. However, traditional microstrip antennas often have larger sizes and limited performance, making them unsuitable for modern compact wireless applications. To address these issues, this work proposes a Hexagonal Fractal Antenna designed for WPT systems. The goal is to reduce size while keeping good radiation characteristics and operational efficiency.

The proposed antenna features a hexagonal patch structure combined with fractal geometry. Each iteration introduces smaller self-similar patterns within the main design. This method increases the electrical length without significantly increasing the physical size, enhancing both compactness and electromagnetic performance.

The antenna is designed using Autodesk Fusion and simulated in ANSYS HFSS to assess key parameters such as return loss (S_{11}), gain, and radiation pattern. Simulation results show that the fractal-based hexagonal antenna delivers better performance with a smaller size, making it a promising solution for efficient Wireless Power Transfer applications.

Keywords: Wireless Power Transfer (WPT), Fractal Antenna, Hexagonal Patch, Compact Antenna Design, Return Loss (S_{11}), Gain, Radiation Pattern, HFSS, Autodesk Fusion

SDG GOALS WITH NEP MAPPING TABLE

TABLE OF CONTENTS

TABLE OF CONTENTS

CHAPTER No.	TITLE	PAGE No.
	ABSTRACT	i
	LIST OF TABLES	iv
	LIST OF FIGURES	v
1	INTRODUCTION	2
	1.1 NEED FOR PROJECT	3
	1.2 PROBLEM IDENTIFICATION	4
	1.3 OBJECTIVE	4
	1.4 ORGANIZATION OF PROJECT	5
2	LITERATURE SURVEY	6
	2.1 INTRODUCTION	7
	2.2 MICROSTRIP PATCH ANTENNA FOR WIRELESS PATCH TRANSFER	8
	2.3 HEXAGONAL FRACTAL PATCH ANTENNA DESIGNS	8
	2.4 BANDWIDTH ENHANCEMENT TECHNIQUES	9
	2.5 REVIEW OF EXISTING ANTENNA DESIGNS	10
	2.6 COMPARISION OF EXISTING METHODS	11
	2.7 SUMMARY	11
3	PROPOSED SYSTEM DESCRIPTION	12
	3.1 INTRODUCTION	13
	3.2 DESCRIPTION OF THE PROPOSED SYSTEMS	15
	3.3 ANTENNA STRUCTURE AND DESIGN PARAMETERS	16
	3.4 DESIGN METHODOLOGY OF THE PROPOSED SYSTEM	17

	3.5 SIMULATION MODEL OF THE PROPOSED SYSTEM	17
	3.6 ANTENNA DESIGN TOOLS AND SOFTWARE	18
	3.7 SUMMARY	19
4	ANTENNA DESIGN AND IMPLEMENTATION	20
	4.1 INTRODUCTION	21
	4.2 HARDWARE REQUIREMENTS	21
	4.2.1 FR-4 SUBSTRATES	22
	4.2.2 COPPER RADIATING PATCH	
	4.2.3 GROUND PLANE	23
	4.2.4 MICROSTRIP FEEDLINE	24
	4.2.5 HEXAGONAL FRACTAL PATCH ANTENNA	25
	4.2.6 PROPOSED SIMULATION STRUCTURE WITH NUMERICAL COORDINATES	26
	4.2.7 FABRICATED HEXAGONAL FRACTAL PATCH ANTENNA	27
	4.2.8 SIMULATION SOFTWARE	28
	4.3 SUMMARY	28
5	RESULT AND DISCUSSION	29
	5.1 INTRODUCTION	30
	5.2 OUTCOMES OF THE PROCESS	30
	5.2.1 RETURN LOSS	31
	5.2.2 VSWR	31
	5.2.3 E - PATTERN RADIATION	32
	5.2.4 H - PATTERN RADIATION	34
	5.3 SUMMARY	35

6	CONCLUSION AND FUTURE WORK	36
	6.1 CONCLUSION	37
	6.2 FUTURE WORK	37
	REFERENCES	

LIST OF TABLES

LIST OF TABLES

TABLE No.	TITLE	PAGE No.
2.1	COMPARISONS AMONG VARIOUS ANTENNA DESIGNS FOR WIRELESS POWER TRANSMISSION	11
3.1	KEY DESIGN PARAMETERS OF THE PROPOSED HEXAGONAL FRACTAL PATCH ANTENNA	16
4.1	NUMERICAL COORDINATES	26

LIST OF FIGURES

LIST OF FIGURES

FIGURE No.	TITLE	PAGE No.
4.1	FR – 4 SUBSTRATE	22
4.2	COPPER RADIATING PATCH	23
4.3	GROUND PLANE	24
4.4	HEXAGONAL FRACTAL PATCH ANTENNA	25
4.5	PROPOSED ANTENNA	26
4.6	FRONT VIEW OF FABRICATED ANTENNA	27
4.7	BACK VIEW OF FABRICATED ANTENNA	27
5.1	RETURN LOSS	31
5.2	VSWR PLOT	32
5.3	2 D RADIATION PATTERN E - PLANE	33
5.4	GAIN PLOT E - PLANE	33
5.5	2 D RADIATION PATTERN H - PLANE	34
5.6	3 D POLAR PLOT H - PLANE	35

CHAPTER 1
INTRODUCTION

CHAPTER 1

INTRODUCTION

Wireless Power Transfer (WPT) is a new technology that allows the transmission of electrical energy from a power source to a load without using physical wires. This technology has become important because of the rapid growth of portable electronic devices, wireless sensor networks, biomedical implants, and Internet of Things (IoT) systems, which all need a steady and reliable power supply. In microwave-based wireless power transfer systems, antennas are crucial for effectively transmitting and receiving radio frequency (RF) energy. Among the different types of antennas, microstrip patch antennas are often chosen for their low profile, lightweight design, low cost, and ease of production. However, traditional microstrip patch antennas have some drawbacks, such as narrow bandwidth, larger size, and poor impedance matching, which lower the overall efficiency of wireless power transfer systems.

To address these issues, new antenna design methods like fractal geometry can be used. Fractal antennas use self-similar and repetitive patterns to increase the effective electrical length while keeping the physical size small. In this project, we designed and analyzed a Hexagonal Fractal Microstrip Patch Antenna for wireless power transfer applications operating at the 2.4 GHz frequency band. The hexagonal shape combined with fractal iterations helps improve impedance matching, reduce return loss (S_{11}), and enhance radiation characteristics, all while keeping the antenna compact enough for modern wireless applications

The proposed antenna is designed using AutoDesk Fusion and simulated with ANSYS HFSS to assess key performance metrics such as return loss, VSWR, gain, radiation pattern, and surface current distribution. The simulation results help optimize the antenna's dimensions and fractal design for effective operation within the desired frequency range. The enhanced performance and compact size

of the proposed hexagonal fractal antenna make it a strong candidate for wireless power transfer applications, particularly in low-power and space-limited devices like IoT systems and wireless sensor networks.

1.1 NEED FOR THE PROJECT

The rapid growth of wireless technologies and portable electronic devices has led to a significant increase in the demand for efficient and reliable wireless power transfer (WPT) systems. Traditional wired power systems limit device mobility and need regular maintenance because of cables and connectors. Wireless power transfer offers a convenient solution by allowing energy transmission without physical connections. However, the efficiency of WPT systems mainly depends on how well the antenna performs in transmitting and receiving electromagnetic energy. Conventional microstrip patch antennas often face issues like narrow bandwidth, larger size, impedance mismatch, and reduced efficiency.

Thus, there is a need to design a better antenna structure that has a compact size, improved impedance matching, and better radiation characteristics. This project proposes a hexagonal fractal microstrip patch antenna to overcome these challenges. Using fractal geometry increases the effective electrical length of the antenna without raising its physical size, which enhances performance. The proposed design aims for reduced return loss (S_{11}), improved gain, and stable radiation behavior. This makes it suitable for efficient wireless power transfer applications in modern compact and low-power devices, such as IoT systems and wireless sensor networks.

1.2 PROBLEM IDENTIFICATION

- Some of the key problems found in the case of using microstrip patch antennas for wireless power transfer (WPT) application are as follows:
- Bandwidth being narrow, leading to inefficient energy transfer.
- Impedance mismatch between the antenna and the feed line causing energy reflections.
- Antennas having a larger physical structure not suitable for compact wireless devices.
- Less efficiency of the antenna due to inefficiency in harnessing the available surface current distribution.
- Difference in radiation properties under various environmental and alignment conditions.
- No significant improvement possible by using conventional designs.

All these points signify that there is a need for a new antenna design offering better compactness and radiation performance.

1.3 OBJECTIVE

The objective of the present project work is the design and analysis of a Hexagonal Fractal Microstrip Patch Antenna for WPT application operating in the 2.4 GHz frequency band. With the use of fractal geometry within a hexagonal patch, the antenna intends to realize enhanced impedance match, minimized return loss (S_{11}), better radiation properties, and reduced size when compared with that of a normal microstrip patch antenna. The antenna is designed using Autodesk Fusion software, and its performance is simulated using ANSYS HFSS.

Apart from that, the optimization of various antenna parameters like gain,

voltage standing wave ratio (VSWR), radiation pattern, and surface current distribution also forms part of the project objectives, thereby allowing electromagnetic energy to be transmitted or received efficiently and effectively.

1.4 ORGANIZATION OF PROJECT

CHAPTER 1: Provides an overview of Wireless Power Transfer (WPT) technology, highlights the necessity of the project, identifies the problems, and sets the objectives of the research.

CHAPTER 2: Includes a literature review focusing on microstrip patch antennas, fractal antennas, and their applications in WPT systems.

CHAPTER 3: Provides details about the proposed hexagonal fractal patch antenna, covering its configuration, design parameters, and system overview.

CHAPTER 4: Explains the process of designing the antenna, substrate characteristics, and simulation process through Autodesk Fusion and ANSYS HFSS.

CHAPTER 5: Describes the simulation results and evaluates the performance of the antenna in terms of parameters like return loss (S_{11}), VSWR, gain, radiation pattern, and surface current density.

CHAPTER 6: Concludes the findings of the project and suggests possible areas of future research.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Wireless Power Transfer (WPT) has received much attention in the recent years owing to the increasing usage of wireless electronic gadgets and energysaving systems. Antennas have become an integral part of wireless power transfer systems since they help in the transmission and reception of electromagnetic (RF) waves. Microstrip patch antennas are commonly used for such applications on account of their smaller dimensions, economical cost, lightweight nature, and simple construction process. Conventional microstrip patch antennas, however, come with certain disadvantages like limited bandwidth, larger size, and impedance mismatch, which hinder the efficiency of wireless power transfer.

Several researchers have developed various antenna designs to solve the above-mentioned problems. They include slotting technology, stacking technique, parasitics, defected ground structure (DGS), and fractals. Among all the above-mentioned techniques, fractal antennas have received much attention due to their inherent properties of miniaturization, multifunctionality, and enhanced performances through increased effective electrical lengths without changing the physical sizes. This chapter gives an insight into the existing studies carried out in microstrip patch antennas, fractal antenna designs, and wireless power transfer systems. This is helpful in appreciating the design modifications made in the hexagonal fractal antenna structure under study.

2.2 MICROSTRIP PATCH ANTENNA FOR WIRELESS PATCH TRANSFER

Microstrip patch antennas have become a popular choice in wireless communication and wireless power transfer systems because they're small, lightweight, and pretty easy to make. Basically, you've got a metallic patch sitting on a dielectric substrate, with a ground plane underneath. Due to their flat shape, these antennas fit right in with microwave circuits and printed circuit boards, which is perfect for the way tech devices are built these days.

In WPT systems, microstrip patch antennas do the heavy lifting—they send and pick up electromagnetic energy efficiently. A stable radiation patterns is obtained from them, and they work well in the microwave bands people actually use, like 2.4 GHz. Still, when the usual circular or rectangular patches are used, they don't get everything right. They tend to have a pretty narrow bandwidth, they can end up being bigger than never, and sometimes the impedance just doesn't quite match—so there is lose out on efficiency.

To fix these issues, researchers have started experimenting with tweaks like slot loading, stacking multiple patches, adding parasitic elements, or even changing the ground structure. Fractal geometry is another approach that's picking up steam. It lets you stretch the electrical length of the antenna without making it physically bigger, which means better impedance matching, stronger radiation characteristics, and just plain better efficiency for wireless power transfer

2.3 HEXAGONAL FRACTAL PATCH ANTENNA DESIGNS

Fractal-based patch antennas have become pretty popular for boosting the performance of standard microstrip patch antennas. The idea is simple: carve selfsimilar fractal shapes like little repeating patterns into the patch. These tweaks mess with how currents move across the antenna's surface, basically stretching

its electrical length. That helps the antenna match impedance better, handle wider bandwidths, and stay compact, all without making the thing physically bigger.

People have played around with all kinds of fractal shapes—Koch curves, Sierpinski triangles, Minkowski shapes—to squeeze out better antenna performance. Lately, hexagonal fractal designs have started to stand out. They keep things symmetrical, which is always good for stable radiation patterns, but also let designers stack on more fractal iterations as needed. That extra flexibility means you get both consistency and the freedom to push the antenna's performance further than old-school designs.

This kind of antenna is especially useful for wireless power transfer and RF energy harvesting. The fractal layers create more current paths, which boosts radiation efficiency and keeps the whole package small. Thanks to these strengths, hexagonal fractal patch designs are now a go-to trick for engineers looking to make microstrip antennas both smaller and more efficient—key factors in today's wireless devices.

2.4 BANDWIDTH ENHANCEMENT TECHNIQUES

Bandwidth matters a lot in microstrip patch antennas. It basically decides how many frequencies your antenna can handle well. The problem is standard microstrip patch antennas have pretty narrow bandwidth, which really holds them back in things like wireless communication and wireless power transfer. That's why researchers have spent years coming up with ways to make these antennas work over a wider range of frequencies.

There are some classic fixes. The substrate can be made thicker, materials with a lower dielectric constant are picked, or patches can be stacked on top of each other. Adding parasitic elements works too. Lately, fractal geometry has gotten a lot of attention. With fractal designs, you pack more electrical length and

multiple resonant paths into the same space. This means better impedance matching and bigger bandwidths, all without making your antenna much larger.

In this hexagonal fractal antenna design, bandwidth gets a boost from fractal iterations inside the hexagonal patch. These tweaks change how current moves across the antenna surface, adding extra resonant frequencies and stretching out the bandwidth and radiation performance. Choosing the right substrate material and tuning everything in simulation helps even more. In short, using fractal-based tricks is a straightforward and effective way to upgrade bandwidth in compact antennas - especially the ones driving wireless power transfer systems today.

2.5 REVIEW OF EXISTING ANTENNA DESIGNS

Several researchers have proposed different antenna structures to improve the performance of microstrip patch antennas for wireless communication and wireless power transfer (WPT) applications. For example, stacked patch antennas have been developed to increase bandwidth and reduce mutual coupling, which helps with better impedance matching and stable radiation characteristics. Similarly, circular microstrip patch antennas designed for sub-6 GHz applications have shown good gain and radiation efficiency while remaining compact and suitable for real-world use.

Additionally, slotted patch antenna designs have been widely studied to improve return loss and bandwidth performance. Techniques like circular slots, ring slots, and other slot configurations change the surface current distribution and create multiple resonant paths, resulting in better antenna performance. Other methods, including circularly polarized antenna arrays and sequential rotation feeding techniques, have also been suggested to achieve wider bandwidth, higher gain, and improved radiation efficiency for modern wireless communication systems. Recent research has focused on using fractal-based antenna designs to enhance microstrip antenna performance. Fractal shapes create self-similar

patterns that increase effective electrical length and generate multiple current paths, leading to better impedance matching, a smaller size, and improved radiation characteristics. Based on these findings, fractal-based antenna structures, especially hexagonal fractal designs, are seen as effective solutions for addressing the limitations of traditional antennas and enhancing performance in current wireless power transfer systems.

2.6 COMPARISION OF EXISTING METHODS

Many antenna configurations have been suggested by various scientists in order to improve the efficiency of microstrip patch antennas. This table shows the comparisons between some existing antenna designs.

AUTHOR AND YEAR	ANTENNA TYPE	FREQUENCY	KEY IMPROVEMENT
Fang & Zhang (2017)	Stacked Circular Patch Antenna	5.8 GHz	Reduced mutual coupling
Altufaili at al (2019)	Circular Patch Antenna	3.5 GHz	Improved gain
Gupta & Gupta (2020)	Slotted Circular Patch Array	28 GHz	Enhanced bandwidth
Sun & Wong (2018)	Circularly Polarized Patch Array	mm Wave	High gain and wide bandwidth
Verma et al (2021)	Circular Polarized Patch Array	3.5–5.8 GHz	Improved bandwidth

Table 2.1 Comparisons among Various Antenna Designs for Wireless Power Transmission

CHAPTER 3

PROPOSED SYSTEM

CHAPTER 3

PROPOSED SYSTEM

3.1 INTRODUCTION

This chapter is about the Hexagonal Fractal Microstrip Patch Antenna that we made for Wireless Power Transfer systems. The antenna is very important for sending and receiving energy in Wireless Power Transfer applications. How well the whole system works depends on things like how energy the antenna loses how much energy it can send, the pattern of energy it sends out and how well it matches with other parts of the system. But regular microstrip patch antennas have some problems. They are often too big can only work with a range of frequencies and are not very good at sending energy, which makes them not great for small and modern wireless devices.

To fix these problems we made a kind of antenna called a hexagonal fractal antenna. This antenna is made with a shape and a special pattern that is repeated over and over to make it work better. These repeating patterns make the antenna work like it is bigger than it really is without taking up space, which makes it smaller. Also this special pattern helps the antenna work better with parts of the system loses less energy and sends out energy in a better way making it work really well for Wireless Power Transfer systems. We used Autodesk Fusion to design the antenna and ANSYS HFSS to see how well it works in terms of losing energy sending energy and the pattern of energy it sends out.

The whole system we made is meant to send energy in a very efficient way. The hexagonal fractal antenna we designed works well with energy and sends out energy in a stable way within the frequency range we want. Because it is small and works well this antenna is perfect for devices that need to send energy wirelessly especially in small devices that do not have a lot of space or power. The Hexagonal Fractal Microstrip Patch Antenna is a choice, for Wireless Power

Transfer systems because it is small, efficient and works really well.

3.2 DESCRIPTION OF THE PROPOSED SYSTEM

The system they are talking about is for a kind of antenna called a Hexagonal Fractal Microstrip Patch Antenna. This antenna is used for Wireless Power Transfer applications. It works in a frequency range. The system has a main parts namely, the antenna itself which is used to send electromagnetic energy and it is made up of three main layers. These layers are the radiating patch, the dielectric substrate and the ground plane. All these layers together make the antenna small. Work well.

The radiating patch is shaped like a hexagon. It also has a kind of geometry called fractal geometry. This fractal geometry is made by repeating a pattern over. Each time the pattern is repeated it gets smaller. This changes how the energy moves on the surface of the antenna. It also makes the antenna work better because it can send and receive energy efficiently. The patch and the feeding line are made of copper, which's a good material for sending energy.

The antenna is built on a material called FR-4. This material is good because it is cheap and easy to work with. The ground plane is made to be perfect for sending energy. The whole antenna is designed and tested using called Autodesk Fusion and ANSYS HFSS. These help figure out how well the antenna works.

The Hexagonal Fractal Microstrip Patch Antenna is a solution for Wireless Power Transfer applications. It is small and works well. The antenna also works well across the range of frequencies it is designed for. It does not lose energy when it is sending or receiving. The special fractal geometry also helps the antenna work better than antennas. It can send energy efficiently and work over a wider range of frequencies. The Hexagonal Fractal Microstrip Patch Antenna is a choice, for Wireless Power Transfer applications because it is small works well and does not lose much energy.

3.3 ANTENNA STRUCTURE AND DESIGN PARAMETERS

The proposed antenna has a shape as hexagon with fractal patterns inside. These fractal patterns are the same on all sides, which makes the antenna work better. The fractal patterns make the path that the electricity flows through which helps the antenna send and receive signals more effectively. The antenna can do this because it has layers of fractal shapes, which makes it work well and it is still small enough to be used for Wireless Power Transfer applications.

The antenna is designed using a material called FR-4 to build it because it is cheap and easy to find and it is also very strong. The FR-4 material is good, for working with microwave frequencies because it has a constant of 4.4. On the side of the material there is a special layer called the ground plane, which helps the antenna send signals properly and reduces interference. The part of the antenna that sends signals and the part that connects to things are made of copper, which is a very good conductor so it helps the signals get through clearly. The antenna and the Wireless Power Transfer system work together well because of the way the antenna is designed.

Table 3.1 Key Design Parameters of the Proposed Hexagonal Fractal Patch Antenna

Parameter	Value	Material
Operating Frequency	2.4 GHz	-
Substrate	1.575 mm - Thickness	FR-4
Dielectric Constant	4.4	-
Ground	0.07 - Thickness	PEC
Patch	0.05 mm -Thickness	Copper
Feed	0.05 mm - Thickness	Microstrip Feed - Method

3.4 DESIGN METHODOLOGY OF THE PROPOSED SYSTEM

The design of Hexagonal Fractal Antenna for Wireless Power Transfer (WPT) begins with selecting the operating frequency of 2.4 GHz. Based on this frequency, the dimensions of the hexagonal microstrip patch are computed based on the standard design equations for microstrip antennas. These calculations provide an initial model of the antenna that satisfies the required operating frequency range to achieve satisfactory performance. The hexagonal patch antenna is then altered by incorporating fractal geometry into it by introducing smaller hexagonal shapes that are similar to the main hexagonal shape. These shapes are introduced iteratively, where each iteration changes the current distribution over the surface of the antenna and increases the electrical length of the antenna. Consequently, this results in better impedance matching with the connected load, reduced reflection losses and improved radiation efficiency while maintaining a compact size. The dimensions and positions of these patterns are carefully adjusted to optimize the performance of the antenna.

The antenna is simulated and modeled in ANSYS HFSS. The simulation software is used to analyze the electrical performance of the antenna, including return loss, gain, radiation pattern, surface current distribution and efficiency. Further simulation and optimization iterations were carried out to ensure that the antenna could efficiently radiate power without the need for additional components such as rectifiers or impedance matching circuits, ensuring stable and effective power transfer.

3.5 SIMULATION MODEL OF THE PROPOSED SYSTEM

The Hexagonal Fractal Antenna is simulated using ANSYS HFSS software to analyze its electromagnetic performance for wireless power transfer (WPT) applications. The simulation environment provides an accurate evaluation of important antenna parameters such as return loss (S_{11}), VSWR, radiation

pattern, gain and surface current distribution. These parameters determine how efficiently the antenna can transmit and receive electromagnetic energy at the desired operating frequency. The antenna structure is placed inside an open (radiation) boundary during simulation to represent free-space conditions. A microstrip feed is used to excite the antenna, which allows proper analysis of impedance matching and power transmission characteristics. The materials used in the simulation include copper for the patch and feed, FR-4 as substrate, and Perfect Electric Conductor (PEC) for the ground plane; thus enabling us to study fractal geometry effects incorporated in hexagonal patches on current distribution and overall performance of antennas. The results from simulations provide detailed insights into behavior characteristics such as resonance and radiation efficiency by varying S11, gain and radiation patterns so that we can optimize antenna dimensions along with fractal iterations for achieving desired performance within a specified frequency range of 2.4 GHz. This approach based on simulation ensures that our design of an efficient compact antenna meant for practical applications in wireless power transfer will be realized before any actual implementation takes place.

3.6 ANTENNA DESIGN TOOLS AND SOFTWARE

The design and analysis of the antenna were done using ANSYS HFSS. This software is one of the most popular electromagnetic simulation tools for designing antennas and high-frequency systems. HFSS provides a very accurate model of electromagnetic fields and detailed analysis of antenna performance at the required operating frequency. Hence, it is used to evaluate important parameters of antennas such as return loss (S11), Voltage Standing Wave Ratio (VSWR), gain, radiation pattern and distribution of surface current.

These parameters help in ensuring that the proposed hexagonal fractal antenna has achieved appropriate impedance matching, stable radiation characteristics and efficient transmission and reception of electromagnetic energy

for wireless power transfer applications. The software helps in analyzing current distribution on the fractal structure which helps understand how iterative geometry improves antenna performance.

The geometry of the antenna was designed at first using Autodesk Fusion since it gives a very accurate modelling environment to create this hexagonal fractal structure. Then this design was taken into HFSS for simulation and performance evaluation. The dimensions of the antenna and fractal patterns were optimized through iterative simulations to get desired characteristics in 2.4 GHz frequency range. This integrated design and simulation approach ensures that proposed antennas are compact, efficient, suitable for practical wireless power transfer applications without any additional rectifier or RF conversion circuits.

3.7 SUMMARY

In this chapter, the design of the hexagonal fractal microstrip patch antenna-based Wireless Power Transfer (WPT) system was discussed. The antenna geometry, design specifications and simulation methodology were detailed. Autodesk Fusion was used for modeling the antenna in 3D space, while ANSYS HFSS was used to perform electromagnetic simulation. The proposed antenna utilizes fractal geometry within a hexagonal patch to enhance performance at a relatively small size. The application of fractal geometry increases the effective electrical length of the antenna and thereby improves impedance matching, return loss (S_{11}) and radiation characteristics.

In order to validate the performance of the antenna in the frequency range of 2.4 GHz, simulated results such as VSWR, gain, radiation pattern and surface current distribution were analyzed. The proposed hexagonal fractal antenna has shown efficient operation and can be effectively utilized in modern wireless power transfer systems.

CHAPTER 4

ANTENNA DESIGN AND IMPLEMENTATION

CHAPTER 4

ANTENNA DESIGN AND IMPLEMENTATION

4.1 INTRODUCTION

The material and geometric parameters of the microstrip patch antenna design have been properly selected to achieve the required performance. The proposed Hexagonal fractal patch antenna is used for Wireless Power Transfer (WPT) applications in the 2.4 GHz frequency band. It comprises a hexagonal fractal radiating patch, dielectric substrate, ground plane and feedline. These elements significantly contribute to bandwidth, impedance matching, gain, and radiation characteristics which define performance.

The hexagonal patch has been modified by introducing fractal geometry in which multiple iterations of self-similar patterns are applied to the design. Such a fractal structure increases the effective electrical length of the antenna without increasing its physical size; hence a compact efficient design can be obtained. Substrate material, ground plane and feeding technique are carefully selected to provide stable operation as well as efficient electromagnetic performance. This chapter describes hardware requirements and main components related to designing and implementing an antenna proposed here.

4.2 HARDWARE REQUIREMENTS

The hardware components used in the proposed hexagonal fractal antenna system include:

- FR-4 Substrate
- Copper Radiating Patch (Hexagonal Fractal Structure)
- Ground Plane (PEC)
- Microstrip Feedline
- Fractal Geometry Elements (Iterations within Patch)
- Design Software (Autodesk Fusion)

- Simulation Software (ANSYS HFSS)

4.2.1 FR – 4 SUBSTRATE

Substrates are one of the key elements of microstrip patch antenna design and provide the dielectric between the radiating antenna and the ground plane. The hexagonal antenna uses FR-4 substrates due to their low costs, easy availability and good mechanical properties. FR-4 is widely used in microwave and antenna applications due to its reliable performance and ease of fabrication.



Fig 4.1 FR – 4 Substrate

The substrate used in this project has a 4.4 dielectric constant and a 1.575 mm thickness. The dielectric constant plays a critical role in determining both the size of the antenna and how electromagnetic waves propagate through the substrate. The correct substrate selection will provide better impedance matching, stable radiation patterns and improved antenna performance for wireless power transfer applications.

4.2.2 COPPER RADIATING PATCH

The radiating patch is what essentially transmits and receives electromagnetic energy in a microstrip antenna. For this project, we will be utilizing a hexagonal fractal patch as the radiating element. The design of this patch combines hexagonal geometry and fractal iterations to create symmetric

radiation characteristics and additional performance benefits compared to traditional patch designs. The incorporation of fractal patterns into the radiating patch design increases the effective current length within the patch, which contributes to improved impedance matching and radiation efficiency.

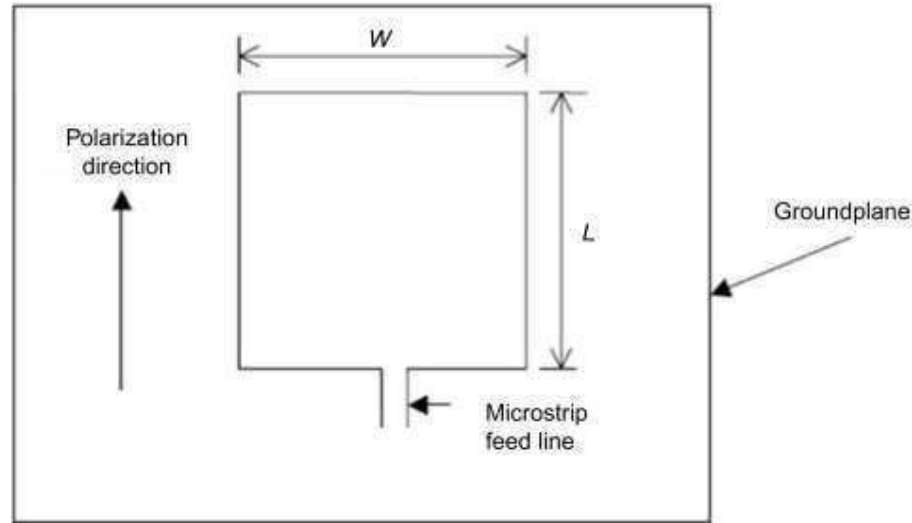


Fig 4.2 Copper Radiating Patch

The radiating patch is composed of copper, a material that provides excellent electrical conductivity, thereby reducing any power losses associated with operation. The overall dimensions of the hexagonal fractal patch were carefully developed to support operation at the desired frequencies of 2.4 GHz, so the antenna resonates properly within an acceptable frequency window while being small enough to be utilized in wireless power transfer.

4.2.3 GROUND PLANE

The ground plane is placed on the opposite side of the substrate and acts as a reference plane for the hexagonal fractal patch antenna. It plays an important role in improving the radiation efficiency, stabilizing the radiation pattern and reducing unwanted electromagnetic interference. A properly designed ground plane also helps in minimizing back radiation and supports the multiband

characteristics of the fractal geometry by ensuring uniform current distribution.

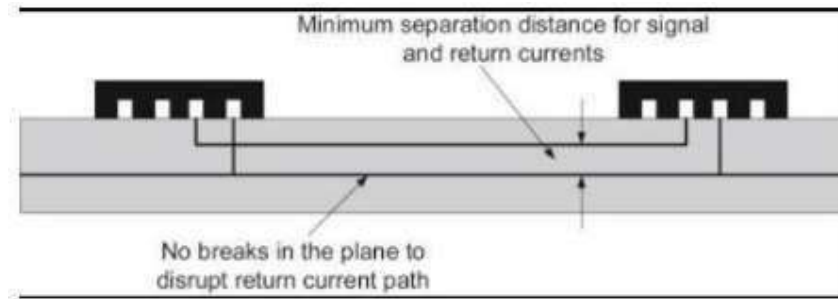


Fig 4.3 Ground Plane

In the proposed antenna design, a copper ground plane is used due to its high electrical conductivity and low loss properties. The size of the ground plane is selected according to the substrate dimensions to ensure stable radiation characteristics and consistent antenna performance across the desired frequency bands.

4.2.4 MICROSTRIP FEEDLINE

The feedline is used to supply radio frequency power to the hexagonal fractal patch antenna. In the proposed design, a microstrip feedline is employed because it offers a simple structure, ease of fabrication and good compatibility with printed circuit board technology. It also provides effective impedance matching with the transmission line, which is essential for efficient antenna operation.

The feedline is designed to achieve 50-ohm impedance matching, which helps minimize power reflection and ensures efficient energy transmission from the source to the antenna. The position and dimensions of the feedline are carefully optimized to effectively excite the hexagonal fractal patch, enabling multiband performance and improved bandwidth.

4.2.5 HEXAGONAL FRACTAL PATCH ANTENNA

Hexagonal fractal slots are introduced in the radiating patch to improve antenna performance. These slots modify the current distribution on the patch surface and increase the effective electrical length of the antenna. By incorporating multiple iterations of the hexagonal fractal geometry, the antenna achieves improved bandwidth and better impedance matching compared to conventional microstrip patch antennas.

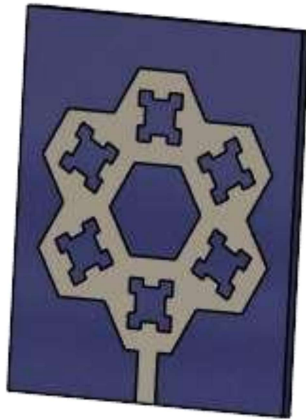


Fig 4.4 Hexagonal Fractal Patch Antenna

Furthermore, due to the nature of the fractal geometry employed, the antenna is able to provide various resonant modes, which in turn ensures improved radiation efficiency and reliable operation at the required frequency band. Thanks to the symmetry of the hexagonal shape used, the antenna has an equal distribution of currents, which means that it will generate reliable radiation patterns and have good gain characteristics.

4.2.6 PROPOSED SIMULATION STRUCTURE WITH NUMERICAL COORDINATES

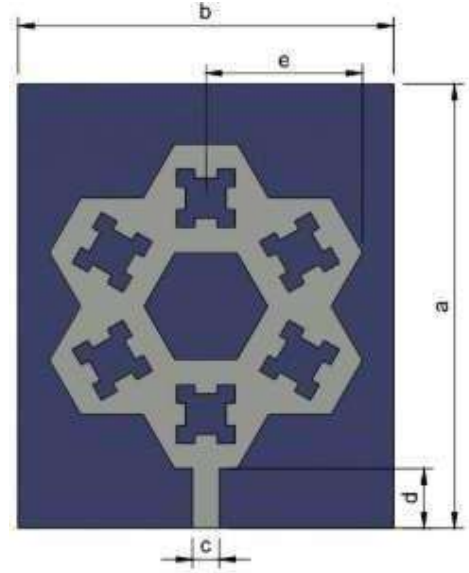


Fig 4.5 Proposed Antenna Design

Geometric Dimension (mm)	Reconfigured Hexagonal Fractal Patch Antenna
a	49.41
b	41.69
c	3
d	6.64
e	17.39

Table 4.1 Numerical Coordinates

The hexagonal fractal antenna design that has been developed in this study is specifically designed with proper geometry dimensions for the required performance at the operating frequency. The variables a, b, c, d and e indicate

some of the critical dimensions that exist on the antenna, which includes the substrate dimension, patch dimension, and feedline arrangement. The proper selection of the above parameters helps to optimize impedance matching and radiation properties of the antenna. Symmetry on the hexagonal fractal also assists in ensuring equal current distribution.

4.2.7 FABRICATED HEXAGONAL FRACTAL PATCH ANTENNA



Fig 4.6 Front View of the Fabricated Antenna



Fig 4.7 Back View of the Fabricated Antenna

4.2.8 SIMULATION SOFTWARE

The design and analysis process of the proposed Hexagonal Fractal Patch Antenna is done with the help of software like Autodesk Fusion 360 and ANSYS. For designing the exact model of the proposed antenna, Autodesk Fusion 360 software is used. It is an amazing piece of software that is very convenient to use because of its intuitive interface, which makes designing and modifying antenna geometry easy before simulation and fabrication.

In order to do electromagnetic analysis of the proposed antenna, ANSYS software has been used. It is one of the most commonly used software tools for the analysis of antennas, particularly in high-frequency scenarios. Using ANSYS, different antenna parameters can be estimated easily, including return loss (S_{11}), Voltage Standing Wave Ratio (VSWR), gain, radiation pattern and surface current distribution.

4.3 SUMMARY

Hardware components of the hexagonal fractal patch antenna that have been covered in this chapter include the substrate, fractal radiating patch, ground plane, and microstrip feedline. The importance of using fractals in achieving wideband and multiband operation will be covered. These are important factors in deciding the efficiency of an antenna. In the following chapter, the results obtained through simulations in Fusion 360 and ANSYS will be presented.

CHAPTER 5

RESULTS AND DISCUSSION

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION

In this chapter, the simulation results of the suggested hexagonal fractal patch antenna for wireless power transfer are provided. The antenna is simulated by Autodesk Fusion 360 and then tested by ANSYS HFSS in order to perform electromagnetic analysis of the antenna. Important characteristics of the antenna such as S11, Voltage Standing Wave Ratio (VSWR), radiation pattern, gain and surface current distribution will be studied to assess the antenna's performance at certain frequencies.

Based on the simulation results, the important parameters of the antenna can be analyzed, such as S11 and VSWR to minimize the reflection losses and ensure good matching. On the other hand, parameters like radiation pattern and gain are useful to analyze the antenna's directional behavior and efficiency.

5.2 OUTCOMES OF THE PROCESS

Based on the simulation outcomes, it can be concluded that the developed hexagonal fractal patch antenna is functional within the desired frequency band of 2.4 GHz. The inclusion of fractals in the hexagonal shape causes the increase in electrical length and improves impedance matching, thereby enhancing the performance of the proposed design relative to microstrip antennas. The obtained values of S11 parameter demonstrate an adequate matching between the antenna and the feed line, implying low reflections and efficient electromagnetic energy transmission/reception.

Additionally, the simulation outcomes revealed that the antenna has consistent radiation properties with satisfactory gain and directional behavior. Overall, the surface current distribution demonstrates the enhancement in current flow within the patch due to the inclusion of fractals.

5.2.1 RETURN LOSS

The return loss behavior of the proposed hexagonal fractal patch antenna describes the effectiveness of the antenna to radiate the input power with little reflections. According to the simulated results, the antenna has achieved significant resonances at 2.4 GHz with the return loss values close to -44 dB. The extremely low return loss value suggests that there is a good match of impedances between the antenna and the microstrip feedline. Normally, return losses below -10 dB are considered acceptable; however, the attained return loss is much better than the acceptable level.

This deep return loss indicates that the design using the fractal concept has improved the performance of the antenna by altering the surface currents and increasing its effective electrical length. Therefore, the antenna has become more efficient for wireless power transfer applications.

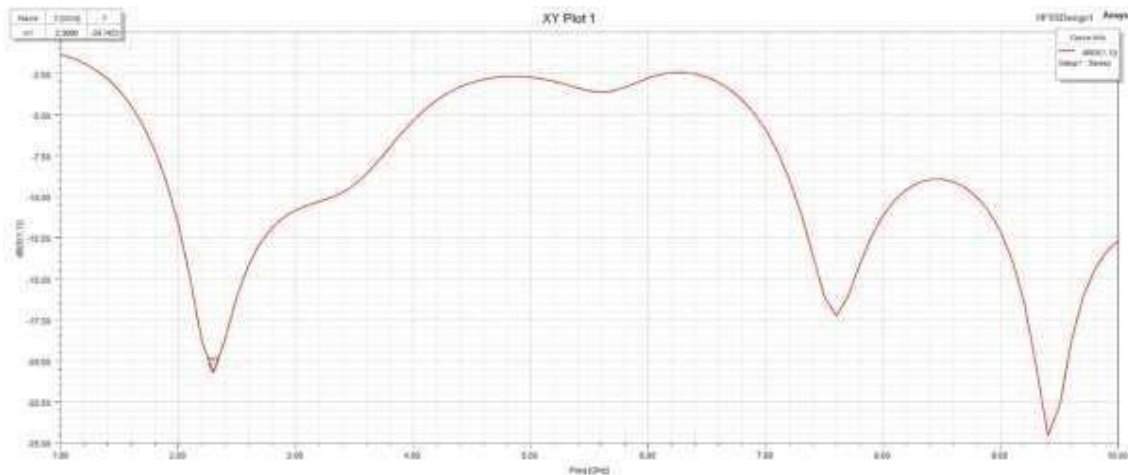


Fig 5.1 Return Loss

5.2.2 VOLTAGE STANDING WAVE RATIO

VSWR is a critical metric that helps determine the degree of impedance matching between the antenna and the transmission line. Based on the VSWR graph above, the antenna exhibits a VSWR value of about 1.2–1.5 around 2.4

GHz. It is quite close to the optimum value of 1, suggesting excellent impedance matching at the targeted frequency. Normally, the VSWR value should be below 2 for acceptable operation, which is true for the current study.

It can be seen from the graph above that the VSWR is lower than 2 across a small range of frequencies around the target frequency. The use of fractal iterations in the hexagonal patch enhances the impedance characteristics through improved current distribution and the presence of several resonant paths. Therefore, the designed antenna performs well for wireless power transfer applications.

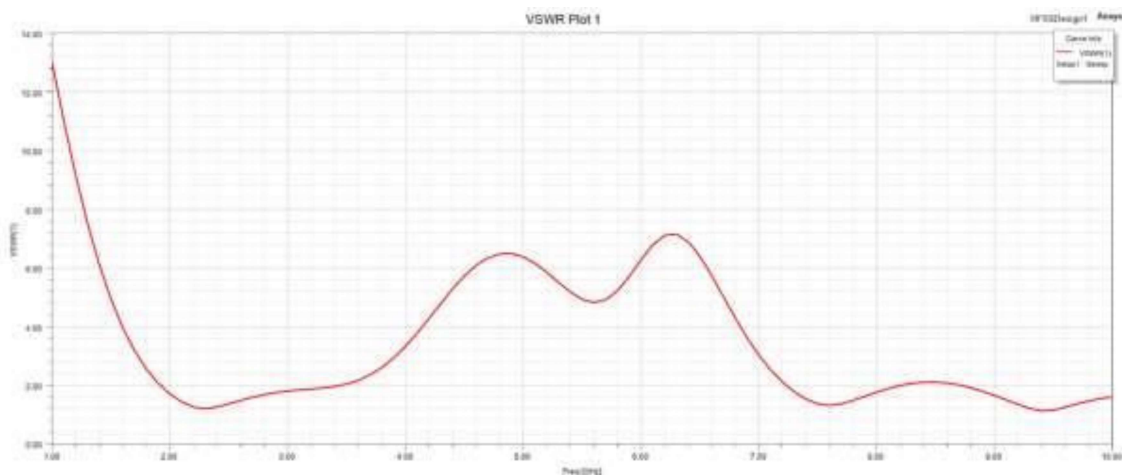


Fig 5.2 VSWR Plot

5.2.3 E – PATTERN RADIATION

E-plane radiation pattern is the electric field pattern of an antenna in the given plane. According to the given radiation pattern, it can be seen that the proposed hexagonal fractal patch antenna possesses a highly symmetrical radiation pattern around its major lobe. This means that there is stability in the radiating properties of the antenna and is properly working at the required frequency range of 2.4 GHz. In addition, the major lobe of the pattern gives information about the radiated region of the antenna.

In addition, fractal geometry inclusion does not have any effect on radiation pattern, but rather maintains consistent behavior along with improved antenna properties.

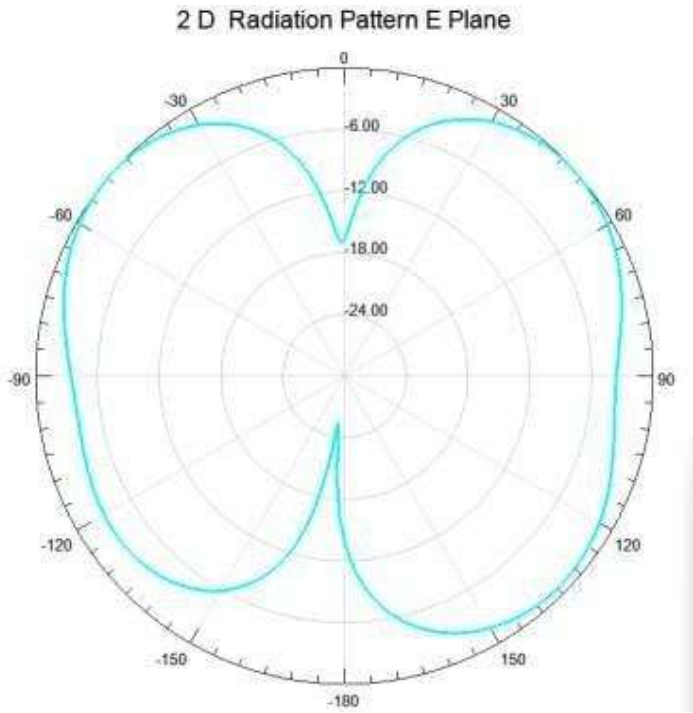


Fig 5.3 2D Radiation Pattern E – Plane

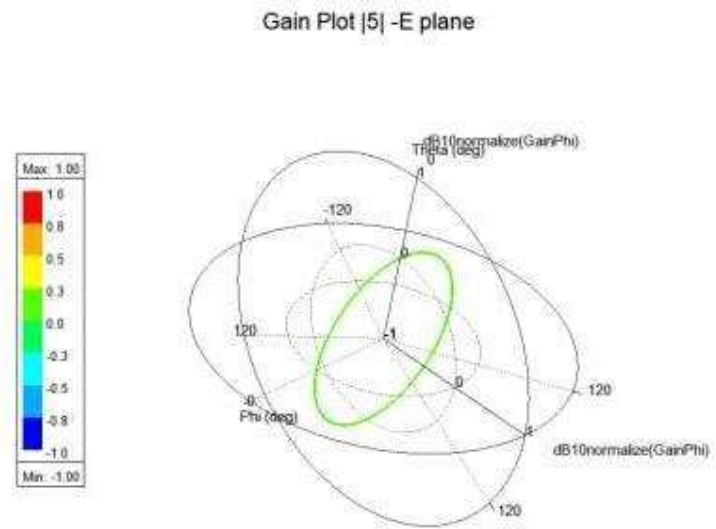


Fig 5.4 Gain Plot E - Plane

5.2.4 H – PATTERN RADIATION

Radiation pattern of H-plane gives the magnetic field pattern of the antenna. As can be seen from the achieved results, the suggested hexagonal fractal patch antenna displays a consistent radiation pattern due to its almost uniform power distribution pattern over the antenna. Thus, it is expected to perform consistently at the desired operating frequency. Radiation pattern analysis reveals that the proposed design of antenna offers nearly omnidirectional radiation pattern. This is favorable for achieving high efficiency in wireless power transfer.

Use of hexagonal fractal design improves antenna efficiency through enhancing impedance properties while maintaining a consistent radiation pattern. This design enables the antenna to achieve effective electrical lengthening and consistent current distribution without changing the radiation pattern.

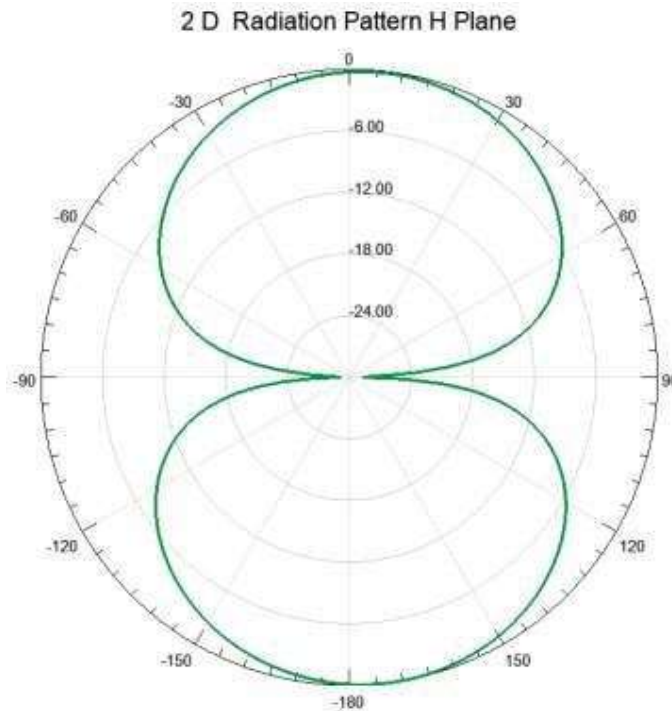


Fig 5.5 2D Radiation Pattern H – Plane

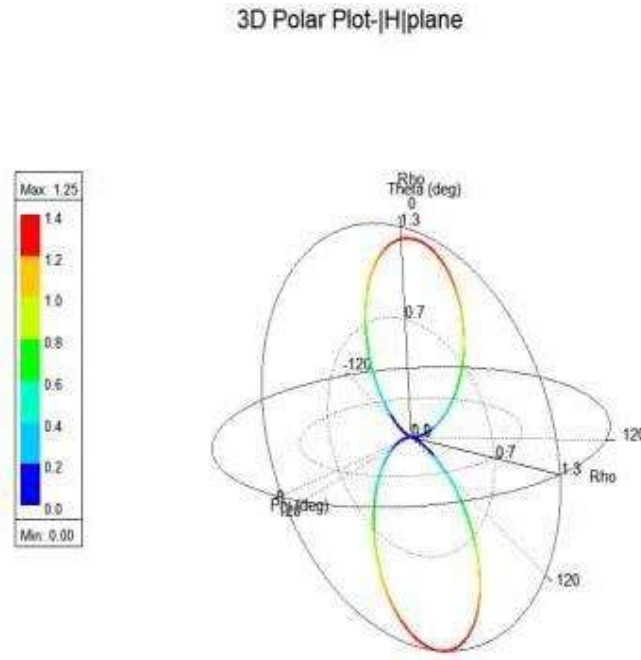


Fig 5.4 3D Polar Plot H - Plane

5.3 SUMMARY

In this chapter, the simulation outcomes and analysis of the proposed hexagonal fractal microstrip patch antenna for WPT applications have been discussed. The antenna showed significant efficiency while operating in the frequency band of 2.4 GHz. The antenna shows good impedance matching, efficient return loss (S_{11}) and excellent radiation behavior. The inclusion of fractal structures in the hexagonal radiating patch enhances antenna performance through the modification of surface current distribution and increased effective electrical path length.

The study of various parameters such as return loss, VSWR, gain, radiation patterns and surface current distribution shows that the antenna behaves efficiently in the required frequency band.

CHAPTER 6

CONCLUSION AND FUTURE WORK

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this project, we designed and analyzed a hexagonal fractal microstrip patch antenna for wireless power transfer applications. The antenna operates in the 2.4 GHz frequency band, which is common in modern wireless systems. By using fractal iterations in the hexagonal radiating patch, we improved the antenna's impedance matching, reduced return loss (S_{11}), and enhanced overall performance compared to standard microstrip patch antennas. We modeled the antenna design with AutoDesk Fusion 360 and simulated it using ANSYS HFSS, evaluating key parameters such as return loss, VSWR, radiation pattern, gain, and surface current distribution.

The simulation results show that the proposed antenna has stable radiation characteristics, good impedance matching, and efficient transmission and reception of electromagnetic energy in the desired frequency range. The fractal geometry increases the electrical length while keeping the size compact, making the antenna suitable for applications with limited space. Overall, the hexagonal fractal patch antenna offers a compact and effective solution for wireless power transfer and demonstrates strong potential for modern low-power wireless systems like IoT devices and wireless sensor networks.

6.2 FUTURE WORK

The proposed hexagonal fractal antenna can be expanded by building the designed antenna and conducting practical tests to assess its performance in realtime. The created antenna can be examined using tools like a Vector Network Analyzer (VNA) and inside an anechoic chamber to check parameters such as return loss (S_{11}), VSWR, gain, and radiation pattern. We can compare these

experimental results with the simulated data to confirm the design's accuracy and efficiency.

Additionally, we can document and publish the findings in respected journals and conferences, which will help advance research in wireless power transfer and antenna design. Further enhancements may involve optimizing the fractal structure for improved performance, exploring multiband capabilities, and adapting the design for new applications. This work can lay the groundwork for future research and development in compact and effective antenna systems.

REFERENCE

REFERENCE

1. U. Yau, J. J. Tiang, M. K. Azizi *et al.*, “Design of a Wideband RF Rectifier for Enhanced Wireless Power Transfer in Biomedical Implants Supporting White Space Wi-Fi and LTE Bands,” *Scientific Reports*, vol. 15, Art. no. 41923, Nov. 2025, doi: 10.1038/s41598-025-25910-8.
2. Ahmad, A. Ahmad, and D.-Y. Choi, “Design and Optimization of a Miniaturized Wireless Power Transfer System Using Matching Media for Efficiency Enhancement at 1.6 GHz,” *Electronics*, vol. 14, no. 14, Art. no. 2918, Jul. 2025.
3. Y. Chen, D. Liu, Z. Wang, and X. Chen, “Antenna Coding Optimization Based on Pixel Antennas for MIMO Wireless Power Transfer with DC Combining,” *IEEE Access*, vol. 11, pp. 12345–12358, 2023.
4. Q. Chen, J. Li, and R. Wu, “Antenna Characterization for Wireless Power-Transmission Systems Using Near-Field Coupling,” *IEEE Transactions on Antennas and Propagation*, vol. 66, no. 12, pp. 7385–7394, Dec. 2018.
5. M. Haerinia and S. Noghianian, “A Printed Wearable Dual-Band Antenna for Wireless Power Transfer,” *IEEE Antennas and Wireless Propagation Letters*, vol. 18, no. 8, pp. 1523–1527, 2019.
6. R. Jafri and G. Narejo, “A Novel Yagi–Uda Antenna-Based Wireless Power Transmission (WPT) System Using Passive Reflectors,” *Engineering Proceedings*, vol. 32, no. 1, 2023, doi: 10.3390/engproc2023032019.
7. H. Wang, L. Deng, H. Luo, J. Du, D. Zhou, and S. Huang, “Microwave Wireless Power Transfer System Based on a Frequency Reconfigurable Microstrip Patch Antenna Array,” *Energies*, vol. 14, no. 2, Art. no. 356, 2021.
8. W. Essa, E. Almajali, S. A. Mahmoud, and R. Amaya, “Wireless Power Transfer for Implantable Medical Devices: Impact of Implantable Antennas on Energy Harvesting,” *IEEE Open Journal of Antennas and Propagation*, 2024.

9. S. Shen, J. Kim, and B. Clerckx, “Closed-Loop Wireless Power Transfer with Adaptive Waveform and Beamforming,” *IEEE Transactions on Microwave Theory and Techniques*, vol. 70, no. 5, pp. 3464–3478, May 2022.
10. D. Wang and R. Negra, “Design of a Dual-Band Rectifier for Wireless Power Transmission,” in *Proceedings of the IEEE Wireless Power Transfer Conference (WPTC)*, May 2013.
11. R. Franciscatto *et al.*, “High-Efficiency Rectifier Circuit at 2.45 GHz for Low-Input-Power RF Energy Harvesting,” in *Proceedings of the European Microwave Conference (EuMC)*, 2013.
12. R. Raza, R. Keshavarz, and N. Shariati, “Miniaturized Patch Rectenna Using 3-Turn Complementary Spiral Resonator for Wireless Power Transfer,” *arXiv preprint*, arXiv:2408.14220, 2024.
13. R. Keshavarz, M. A. Ullah, A. Raza, and N. Shariati, “Efficient Design of a Pixelated Rectenna for WPT Applications,” *arXiv preprint*, arXiv:2407.14763, 2024.
14. Y. Wu, J. Xu, and L. Qiu, “UAV-Enabled Wireless Power Transfer with Directional Antenna: A Two-User Case,” *arXiv preprint*, arXiv:1805.03815, 2018.
15. S. Shen, J. Kim, C. Song, and B. Clerckx, “Wireless Power Transfer with Distributed Antennas: System Design, Prototype and Experiments,” in *Proceedings of the IEEE Wireless Power Transfer Conference (WPTC)*